

Preliminary timeline for the thesis project

Main objective: to design, build, validate and use an automated micro-CT system (computed tomography) to measure swelling in batteries during electrochemical cycling.

Ongoing activities:

- Version control (track all changes, GitHub)
- Documentation and writing in the report.
- Weekly meetings with supervisor (if needed)
- Data backup

Week	Goals	Activities
13-15	Define system design, safety specifications and order necessary components.	• Literature study (battery swelling, CT-based methods etc). • Familiarize myself with Nikon micro-CT and BioLogic SP-50 • Define measurable parameters • Safety analysis (voltage, temperature, emergency stop etc.) • Requirement specification for the holder • Software design • Order components

Deliverables: 1) System block diagram.
2) Safety concept and risk analysis.
3) List of components.

Week	Goals	Activities
16-18	Functional and validated holder.	• CAD-design of the holder • Selection of sensors • Manufacturing and assembly

Deliverables: 1) CAD-files.
2) Assembled holder.
3) Basic functional test.

DECISION: Simplify to manual operation.

Week	Goals	Activities
19-21	Stable experimental workflow.	• Manual CT operation • Experiment protocol definition • Synchronization between CT scans and sensor/potentiostat data

		- Monitoring temperature and voltage - Safety and shutdown logic - Data logging with synchronized timestamps - Simple dashboard - Testing with “dummy” batteries
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Week	STUDY FOR EXAM	STUDY FOR EXAM
22-23	STUDY FOR EXAM	STUDY FOR EXAM

Week	Goal	Activities
24	Stable automated operation.	- API communication (Python) - Control and synchronization of CT-scanning - Monitoring temperature and voltage - Safety and shutdown logic - Data logging with synchronized timestamps - Simple dashboard - Testing with “dummy” batteries

Deliverables: 1) Executable software.
2) Log files.
3) Verified safety behavior.

Week	Goal	Activities
25-28	Validated protocols and improved analysis workflows.	- Preparation of real holders - Testing of charging and scanning sequences - Identification and handling of CT-artefacts - Image registration between cycles - Development of initial swelling curves - Adjustment of experimental parameters

Deliverables: 1) First draft of data.
2) Updated standard procedures.
3) Refined analysis workflows.

Week	Goal	Activities
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29-32	Generate thesis quality data.	• Measurements on at least three cells • Reproducibility analysis • Automatic extraction of swelling measurements • Compilation of results
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Deliverables: 1) Final data.
2) Figures and tables.

Week	Goal	Activities
33-34	Completed thesis project.	• Final data analysis. • Writing and completing the thesis. • Preparing open-source publication.

Deliverables: 1) Completed thesis project.
2) Open-source release.